Sadiq S.

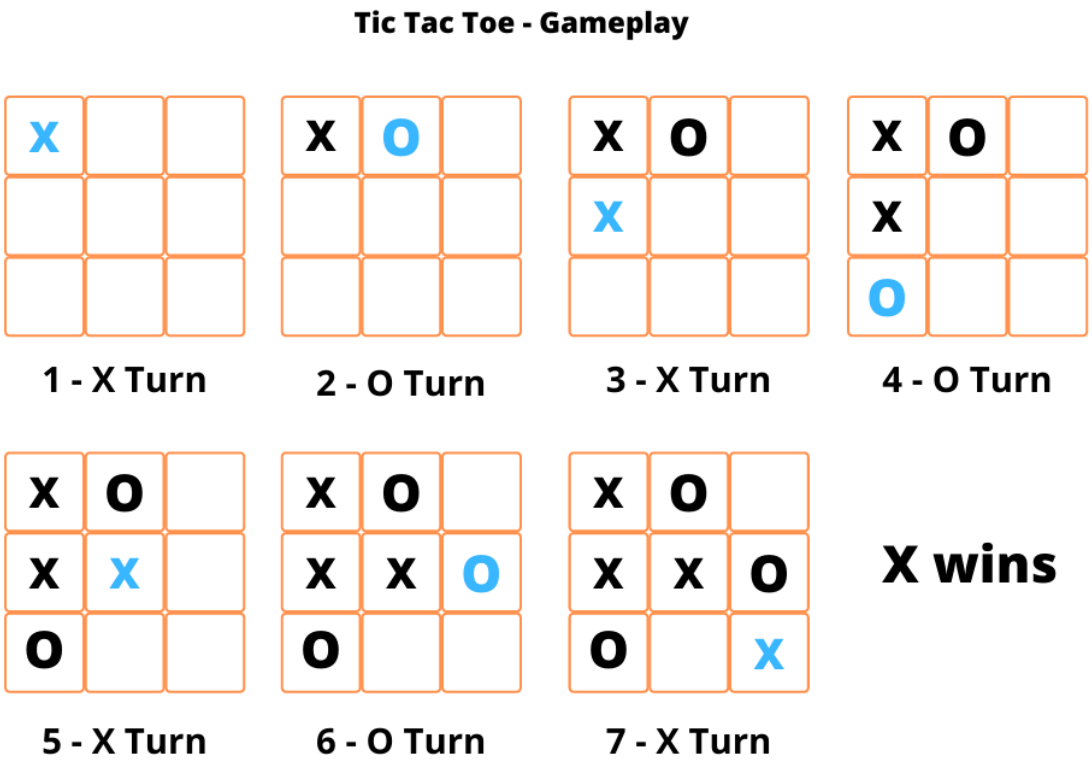
Practical 10

Aim: Demonstrate an IoT based game which can be played between two player who are physically at a considerable distance

Theory:

Rules of the Game:

- The game is to be played between two people.
- One of the player chooses 'O' and the other 'X' to mark their respective cells.
- The game starts with one of the players and the game ends when one of the players has one whole row/ column/ diagonal filled with his/her respective character ('O' or 'X').
- If no one wins, then the game is said to be draw.



- The player X wins the game in the above gameplay. All boxes diagonally fill with X signs. So, the respective player wins the game.

Code:

HTML Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Practical 10</title>
    <link href="https://fonts.googleapis.com/css2?family=Tilt+Warp&display=swap"
rel="stylesheet">
</head>
<style>
    * {
        margin: 0;
        padding: 0;
    }
    body {
        background-color: rgb(176, 156, 200);
        display: grid;
        place-items: center;
        font-family: 'Tilt Warp', cursive;
    }
    h1 {
        margin: 50px 0 50px 0;
        font-size: 5rem;
        color: rgb(0, 50, 50);
        -webkit-text-stroke: .5px black;
    }
    input {
        margin-bottom: 20px;
```

```
    padding: 5px;
    font-size: 1.2rem;
}
button {
    font-size: 1.2rem;
    margin-bottom: 10px;
}
img {
    width: 30px;
}
#cont {
    display: grid;
    grid-template-columns: 1fr 1fr 1fr;
}
.btn {
    font-size: 2rem;
    width: 100px;
    height: 100px;
    cursor: pointer;
    margin: 0;
    background-color: rgb(192, 77, 255);
    border-radius: 10px;
}
#find {
    font-size: 1.5rem;
    color: white;
    cursor: pointer;
    padding: 7px;
    border-radius: 10px;
    width: 250px;
```

```
        background-color: rgb(0, 0, 0);
    }
    .btn:hover {
        background-color: rgb(226, 228, 230);
    }
    button.enabled {
        opacity: 1;
    }
    p {
        font-size: 2rem;
    }
</style>
<body>
    <h1>Tic-Tac-Toe</h1>
    <div style="display: flex; width: 95vw; position: relative;">
        <p id="userCont">You : <span id="user"></span></p>
        <p style="position: absolute; right: 0;" id="oppNameCont">Opponent : <span
id="oppName"></span></p>
    </div>
    <br>
    <p id="valueCont">You are playing as <span id="value"></span></p>
    <br>
    <p id="whosTurn">X's Turn</p>
    <div>
        <p style="font-size: 1.2rem;" id="enterName">Enter your name : </p>
        <input type="text" placeholder="Name" id="name" autocomplete="off">
    </div>
    <button id="find">Search for a player</button>
    <img id="loading" src="loading.gif" alt="">
    <div id="bigcont">
        <div id="cont">
```

```
            <button id="btn1" class="btn"></button>
            <button id="btn2" class="btn"></button>
            <button id="btn3" class="btn"></button>
            <button id="btn4" class="btn"></button>
            <button id="btn5" class="btn"></button>
            <button id="btn6" class="btn"></button>
            <button id="btn7" class="btn"></button>
            <button id="btn8" class="btn"></button>
            <button id="btn9" class="btn"></button>
        </div>
    </div>
</body>
<script src="/socket.io/socket.io.js"></script>
<script>
    document.getElementById("loading").style.display = "none"
    document.getElementById("bigcont").style.display = "none"
    document.getElementById("userCont").style.display = "none"
    document.getElementById("oppNameCont").style.display = "none"
    document.getElementById("valueCont").style.display = "none"
    document.getElementById("whosTurn").style.display = "none"
    const socket = io();
    let name;
    document.getElementById('find').addEventListener("click", function () {
        name = document.getElementById("name").value
        document.getElementById("user").innerText = name
        if (name == null || name == '') {
            alert("Please enter a name")
        }
        else {
            socket.emit("find", { name: name })
```

```
        document.getElementById("loading").style.display = "block"
        document.getElementById("find").disabled = true;
    }
})
socket.on("find", (e) => {
    let allPlayersArray = e.allPlayers
    console.log("html",allPlayersArray)
    if (name != '') {
        document.getElementById("userCont").style.display = "block"
        document.getElementById("oppNameCont").style.display = "block"
        document.getElementById("valueCont").style.display = "block"
        document.getElementById("loading").style.display = "none"
        document.getElementById("name").style.display = "none"
        document.getElementById("find").style.display = "none"
        document.getElementById("enterName").style.display = "none"
        document.getElementById("bigcont").style.display = "block"
        document.getElementById("whosTurn").style.display = "block"
        document.getElementById("whosTurn").innerText = "X's Turn"
    }
    let oppName
    let value
    const foundObject = allPlayersArray.find(obj => obj.p1.p1name == `${name}` ||
obj.p2.p2name == `${name}`);
    foundObject.p1.p1name == `${name}` ? oppName = foundObject.p2.p2name :
oppName = foundObject.p1.p1name
    foundObject.p1.p1name == `${name}` ? value = foundObject.p1.p1value : value =
foundObject.p2.p2value


    document.getElementById("oppName").innerText = oppName
    document.getElementById("value").innerText = value
})
```

```
document.querySelectorAll(".btn").forEach(e => {
    e.addEventListener("click", function () {
        let value = document.getElementById("value").innerText
        e.innerText = value
        socket.emit("playing", { value: value, id: e.id, name: name })
    })
})
socket.on("playing", (e) => {
    const foundObject = (e.allPlayers).find(obj => obj.p1.p1name == `${name}` ||
obj.p2.p2name == `${name}`);
    p1id = foundObject.p1.p1move
    p2id = foundObject.p2.p2move
    if ((foundObject.sum) % 2 == 0) {
        document.getElementById("whosTurn").innerText = "O's Turn"
    }
    else {
        document.getElementById("whosTurn").innerText = "X's Turn"
    }
    if (p1id != '') {
        document.getElementById(`${p1id}`).innerText = "X"
        document.getElementById(`${p1id}`).disabled = true
        document.getElementById(`${p1id}`).style.color = "black"
    }
    if (p2id != '') {
        document.getElementById(`${p2id}`).innerText = "O"
        document.getElementById(`${p2id}`).disabled = true
        document.getElementById(`${p2id}`).style.color = "black"
    }

    check(name, foundObject.sum)
})
```

```
function check(name, sum) {
    document.getElementById("btn1").innerText == '' ? b1 = "a" : b1 =
document.getElementById("btn1").innerText
    document.getElementById("btn2").innerText == '' ? b2 = "b" : b2 =
document.getElementById("btn2").innerText
    document.getElementById("btn3").innerText == '' ? b3 = "c" : b3 =
document.getElementById("btn3").innerText
    document.getElementById("btn4").innerText == '' ? b4 = "d" : b4 =
document.getElementById("btn4").innerText
    document.getElementById("btn5").innerText == '' ? b5 = "e" : b5 =
document.getElementById("btn5").innerText
    document.getElementById("btn6").innerText == '' ? b6 = "f" : b6 =
document.getElementById("btn6").innerText
    document.getElementById("btn7").innerText == '' ? b7 = "g" : b7 =
document.getElementById("btn7").innerText
    document.getElementById("btn8").innerText == '' ? b8 = "h" : b8 =
document.getElementById("btn8").innerText
    document.getElementById("btn9").innerText == '' ? b9 = "i" : b9 =
document.getElementById("btn9").innerText
    if ((b1 == b2 && b2 == b3) || (b4 == b5 && b5 == b6) || (b7 == b8 && b8 == b9) || (b1
== b4 && b4 == b7) || (b2 == b5 && b5 == b8) || (b3 == b6 && b6 == b9) || (b1 == b5 &&
b5 == b9) || (b3 == b5 && b5 == b7)) {
        socket.emit("gameOver", { name: name })
        setTimeout(() => {
            sum % 2 == 0 ? alert("X WON !!") : alert("O WON !!")
            setTimeout(() => {
                location.reload()
            }, 2000)
        }, 100)
    }

    else if (sum == 10) {
        socket.emit("gameOver", { name: name })
        setTimeout(() => {
```

```
            alert("DRAW!!")
            setTimeout(() => {
                location.reload()
            }, 2000)
        }, 100)
    }
  }
</script>
</html>
```

JavaScript Code:

```
const express=require("express")
const app=express()
const path=require("path")
const http=require("http")
const {Server}=require("socket.io")
const server=http.createServer(app)
const io=new Server(server)
app.use(express.static(path.resolve("")))
let arr=[]
let playingArray=[]
io.on("connection",(socket)=>{
  socket.on("find",(e)=>{
    if(e.name!=null){
      arr.push(e.name)
      if(arr.length>=2){
        let p1obj={
          p1name:arr[0],
          p1value:"X",
          p1move:""
        }
```

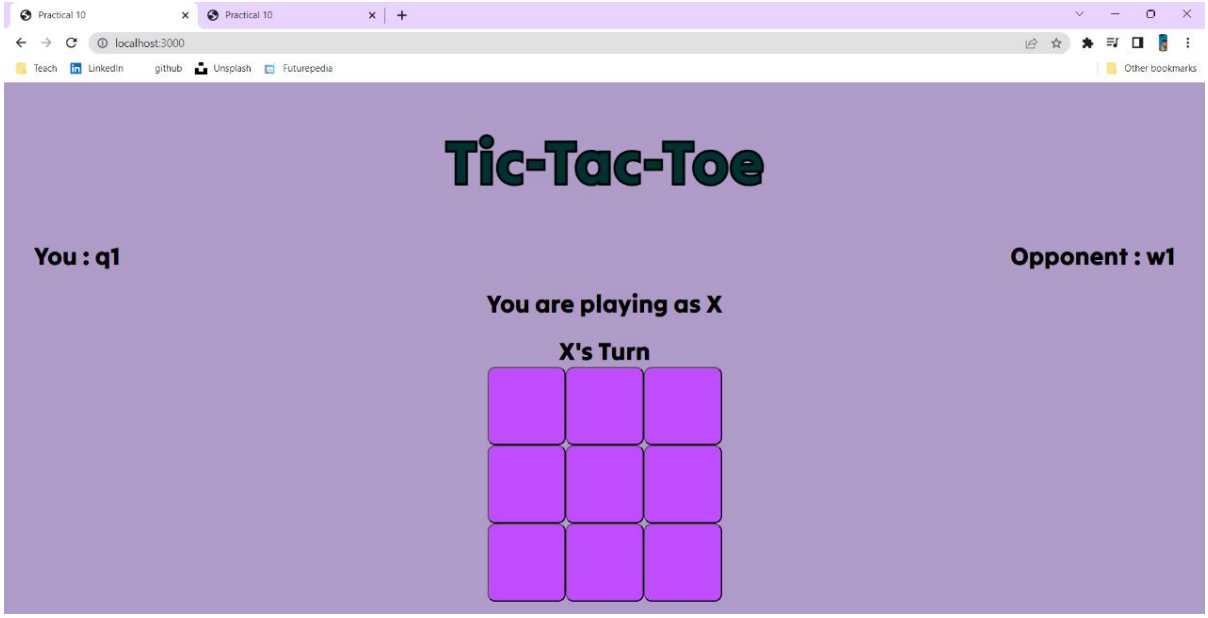
```
            let p2obj={
                p2name:arr[1],
                p2value:"O",
                p2move:""
            }
            let obj={
                p1:p1obj,
                p2:p2obj,
                sum:1
            }
            playingArray.push(obj)
            arr.splice(0,2)
            io.emit("find",{allPlayers:playingArray})
        }
    }
})
socket.on("playing",(e)=>{
    if(e.value=="X"){
        let objToChange=playingArray.find(obj=>obj.p1.p1name===e.name)
        objToChange.p1.p1move=e.id
        objToChange.sum++
    }
    else if(e.value=="O"){
        let objToChange=playingArray.find(obj=>obj.p2.p2name===e.name)
        objToChange.p2.p2move=e.id
        objToChange.sum++
    }

    io.emit("playing",{allPlayers:playingArray})
})
```

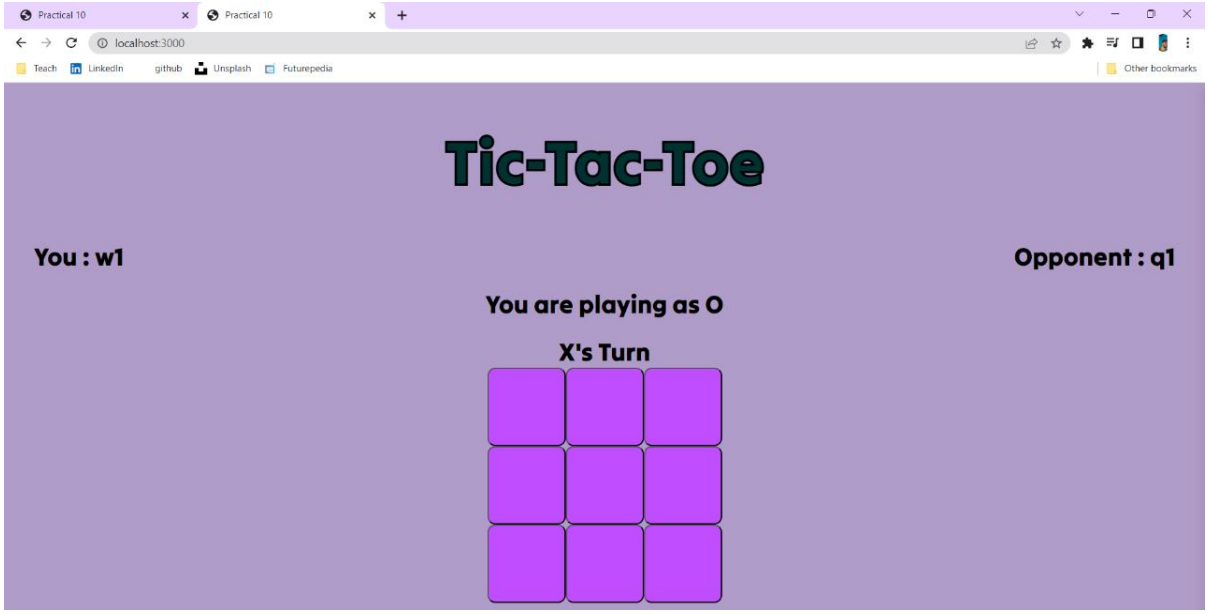
```
    socket.on("gameOver",(e)=>{
        playingArray=playingArray.filter(obj=>obj.p1.p1name!==e.name)
        console.log(playingArray)
        console.log("lol")
    })
})
app.get("/",(req,res)=>{
    return res.sendFile("index.html")
})
server.listen(3000,()=>{
    console.log("port connected to 3000")
})
```

Output:

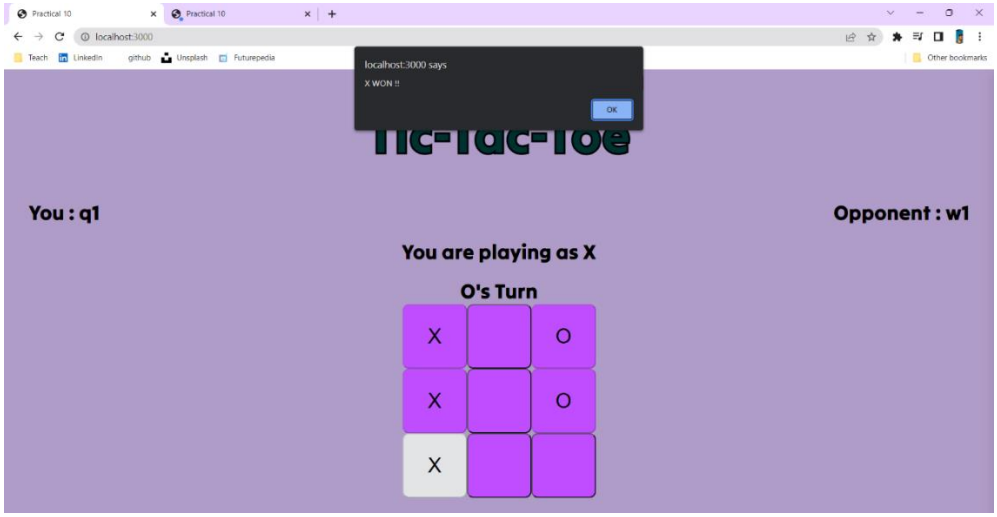
1. Enter Name:

2. Start of the game:



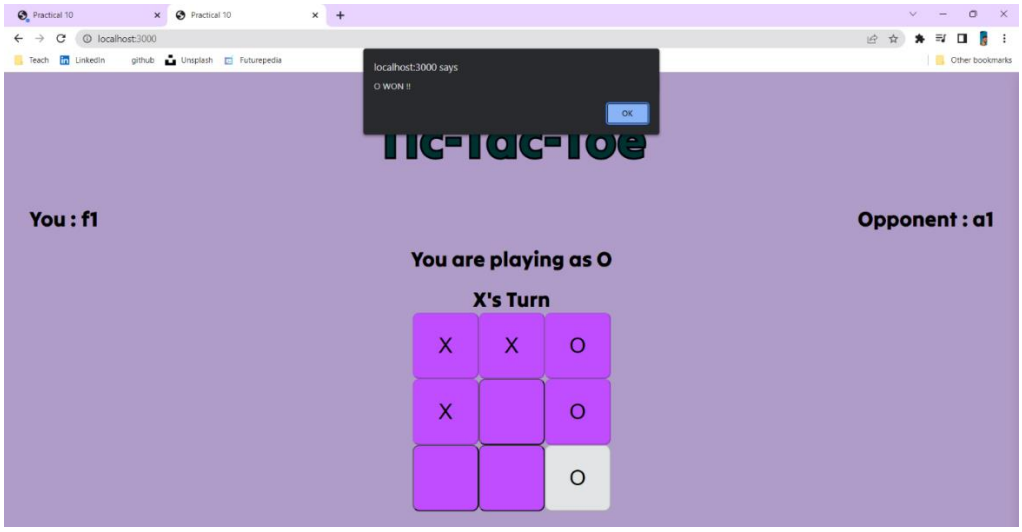


3. When X wins:



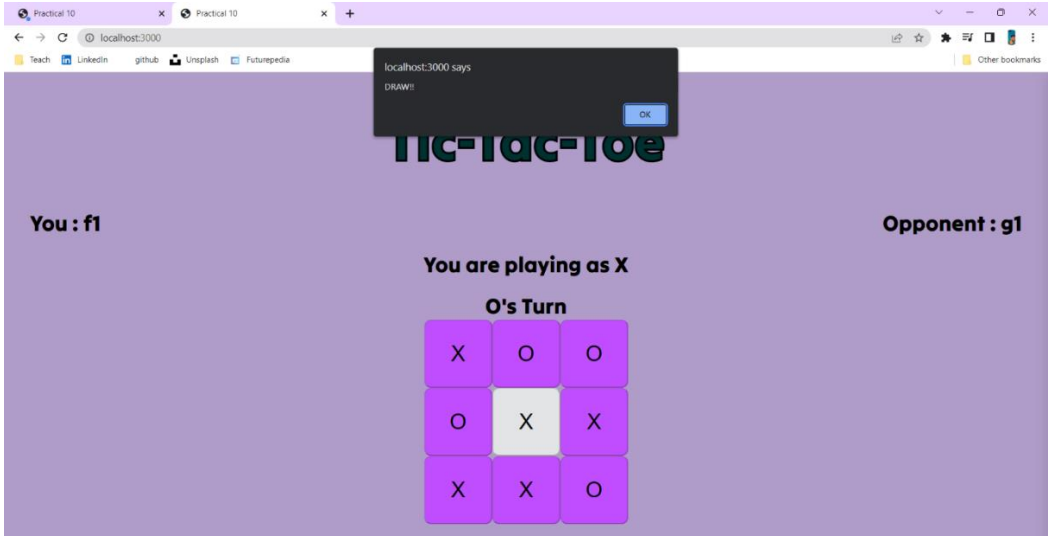
After a player wins, we have to again enter the name

4. When 0 wins:



5. When no one wins the game:

Sadiq S.


Practical 10
Practical 10
localhost:3000
Teach
LinkedIn
github
Unsplash
Futurepedia
Other bookmarks
localhost:3000 says
DRAW!!
OK
Tic-Tac-Toe
You : f1
Opponent : g1
You are playing as X
O's Turn
X O O
O X X
X X O